

Improving Access or Inducing Demand? Analyzing Trade-offs in Healthcare Capacity

Hiroki Saruya

Cabinet Office, Gov. Japan

Masaki Takahashi

Sophia University

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Motivation

- In many countries, governments regulate healthcare capacity
 - Ex) Num. of facilities, beds, MRIs, ...
- **Positive side** : Preventing supplier-induced demand due to excess capacity
 - Capacity constraints can reduce unnecessary care
- **Negative side** : Exacerbating congestion
 - Accessibility in healthcare may decrease
- Prev. literature has studied the positive and negative aspects as separate issues
 - We aim to examine the trade-offs in a unified manner

Overview - Theoretical Part

- Construct an economic model of admissions/discharges in healthcare facility
- Admissions/Discharges depend on bed occupancy through:
 - **Capacity constraint (Congestion)**
 - **Demand inducement**
- Relative importance of these mechanisms is empirically testable

If admission/discharge response to occupancy shock is large

- when occupancy is high → **Capacity constraint** > **Demand inducement**
- when occupancy is low → **Demand inducement** > **Capacity constraint**

Overview - Empirical Part

- Apply the model prediction to **Japanese nursing facilities**
- Exploit **daily patient deaths** in the facility as a source of occupancy shock
→ **Event study & IV estimation**
- What we want to check:
 - How do admissions/discharges respond to occupancy shock?
 - Which mechanism is more important, **capacity constraint** or **induced demand**?

Main Findings

1. Exogenous occupancy decline **increases admissions**, but **little affects discharges**
→ Facilities primarily use admissions to adjust occupancy
2. Admission response is **large** when occupancy is **high**
→ **Capacity constraint** is the main driver of admission response
(especially in high-occupancy facilities)

Policy implication:

Additional capacity in **high-occupancy facilities** can improve access without much supplier-induced demand

Literature

- Selective admissions/discharges based on bed occupancy:
 - He and Konetzka (2015); Gandhi (2022); Corredor-Waldron (2022); Dong et al. (2020); Kim et al. (2015, 2020); Kim and Tong (2022); Freedman (2016); Samiedaluie et al. (2017); Hackmann et al. (2021)
- How to improve healthcare access:
 - Baker and Loyalty (2000); Decker (2007, 2009); Buchmueller et al. (2015); Gandhi(2023); Alexander and Schnell (2024); Cabral et al. (2024)
- **What's new?**
 - Trade-offs between capacity constraints and supplier-induced demand
 - Capacity constraints (congestion) play an important role in admission decisions

Conceptual Framework

Facility Payoffs

- Facility payoffs with an initial num. of patients (occupancy) n , new admissions a , and discharges d :

$$U(n, a, d) = \underbrace{V(rp)}_{\text{income utility}} + \underbrace{b^P p - C^P(p)}_{\text{service utility}} + \underbrace{b^A a - C^A(a)}_{\text{admission utility}} + \underbrace{b^D d - C^D(d)}_{\text{discharge utility}} \quad (1)$$

- $p = n + a - d$: the num. of patients (occupancy) after admission/discharge are realized
- r : per-patient profit
- V converts gross profit rp to utility (**weakly concave**)
- b^P : altruistic per-patient utility from service
- $C^P(p)$: cost of treating p patients (**strictly convex**)
- b^A : altruistic payoffs from admission, C^A : admission cost (**strictly convex**)
- b^D : altruistic payoffs from discharge, C^D : discharge cost (**strictly convex**)

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Avoid low income (occupancy)

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- congestion, binding capacity constraints, ...

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- b^D : altruistic payoffs from discharge, C^D : discharge cost (**strictly convex**)

It is costly to admit or discharge many patients at the same time

Facility Optimal Decisions

- Derive **optimal admissions / discharges** given occupancy n :
 - $(a^*, d^*) = (a^*(n), d^*(n))$
 - $n = \bar{n} \rightarrow \bar{a} = a^*(\bar{n}), \bar{d} = d^*(\bar{n}), \bar{p} = \bar{n} + \bar{a} - \bar{d}$
- Consider how a^* and d^* respond to exogenous decreases in n :
 - $-\frac{\partial a^*}{\partial n}$ and $-\frac{\partial d^*}{\partial n}$
- Assume MC of admissions/discharges are quadratic :
 - $MC^A(a) = \kappa_1^A + \kappa_2^A a$
 - $MC^D(d) = \kappa_1^D + \kappa_2^D d$

$$(\kappa_1^A, \kappa_2^A, \kappa_1^D, \kappa_2^D > 0)$$

Admission/Discharge Response

We can derive the following intuitive results:

(a) **Occupancy**↓ $\Rightarrow$ **Admission**↑, **Discharge**↓

$$-\frac{\partial a^*}{\partial n} > 0 \text{ and } -\frac{\partial d^*}{\partial n} < 0 \quad (\text{The facility tries to fill empty beds})$$

(b) The facility adjusts occupancy in a less costly way

$$\text{Ex) Admission is less costly (small } \kappa_2^A \text{ or large } \kappa_2^D) \rightarrow \left| \frac{\partial a^*}{\partial n}(\bar{n}) \right| \text{ is large}$$

In the following discussion, we focus on admission
(Same results can be applied to discharge)

Decomposition of Admission Response

We can decompose the admission response into two effects:

$$-\frac{\partial a^*}{\partial n} \bigg|_{n=\bar{n}} = \underbrace{-\frac{\kappa_2^D}{D_{J_F}(\bar{a}, \bar{d}; \bar{n})} MB^{A'}(\bar{p})}_{\text{income effect} \geq 0} + \underbrace{\frac{\kappa_2^D}{D_{J_F}(\bar{a}, \bar{d}; \bar{n})} MC^{P'}(\bar{p})}_{\text{cost effect} > 0} \quad (2)$$

- $MB^A(\bar{p})$: Marginal benefit of admission, $MB^{A'}(\bar{p}) \leq 0$, $MB^{A''} \geq 0$
- $MC^P(\bar{p})$: Marginal cost of serving p patients, $MC^{P'}(\bar{p}) > 0$, $MC^{P''} \geq 0$
- $D_{J_F}(\bar{a}, \bar{d}; \bar{n}) > 0$

Income effect : Admit more patients to compensate lost income

Cost effect : Admit more patients due to lower MC of services

supplier-induced demand

easing of capacity constraints

Relative Importance of Cost / Income Effects

$$-\frac{\partial a^*}{\partial n} \bigg|_{n=\bar{n}} = \underbrace{-\frac{\kappa_2^D}{D_{J_F}(\bar{a}, \bar{d}; \bar{n})} MB^{A'}(\bar{p})}_{\text{income effect} \geq 0} + \underbrace{\frac{\kappa_2^D}{D_{J_F}(\bar{a}, \bar{d}; \bar{n})} MC^{P'}(\bar{p})}_{\text{cost effect} > 0}$$

$MB^{A''} \geq 0 \rightarrow$ As occupancy rises, **income effect decreases** } Which change is
 $MC^{P''} \geq 0 \rightarrow$ As occupancy rises, **cost effect increases** } dominant?

The model implies that :

$$\text{If } MC^{P''} > (<) MB^{A''}, \text{ then } \frac{\partial}{\partial n} \left(-\frac{\partial a^*}{\partial n} \right) > (<) 0$$

Admission response is **large** when occupancy is **high** (**low**)

= Admission response is driven by the variation of **cost effect** (**income effect**)

Institutional Background and Data

Nursing Facilities in Japan

- Geriatric Health Services Facility (**GHSF**)
 - **Non-profit** organization
 - Provide **rehabilitative care** for elderly
 - Also provide **end-of-life care** (some patients die in the facility)
- Reimbursement system for GHSF provides attractive setting:
 - **Fixed per-diem payments** account for **90%** = Occupancy is important
 - Payment is risk-adjusted = Cherry-picking is unlikely

Data

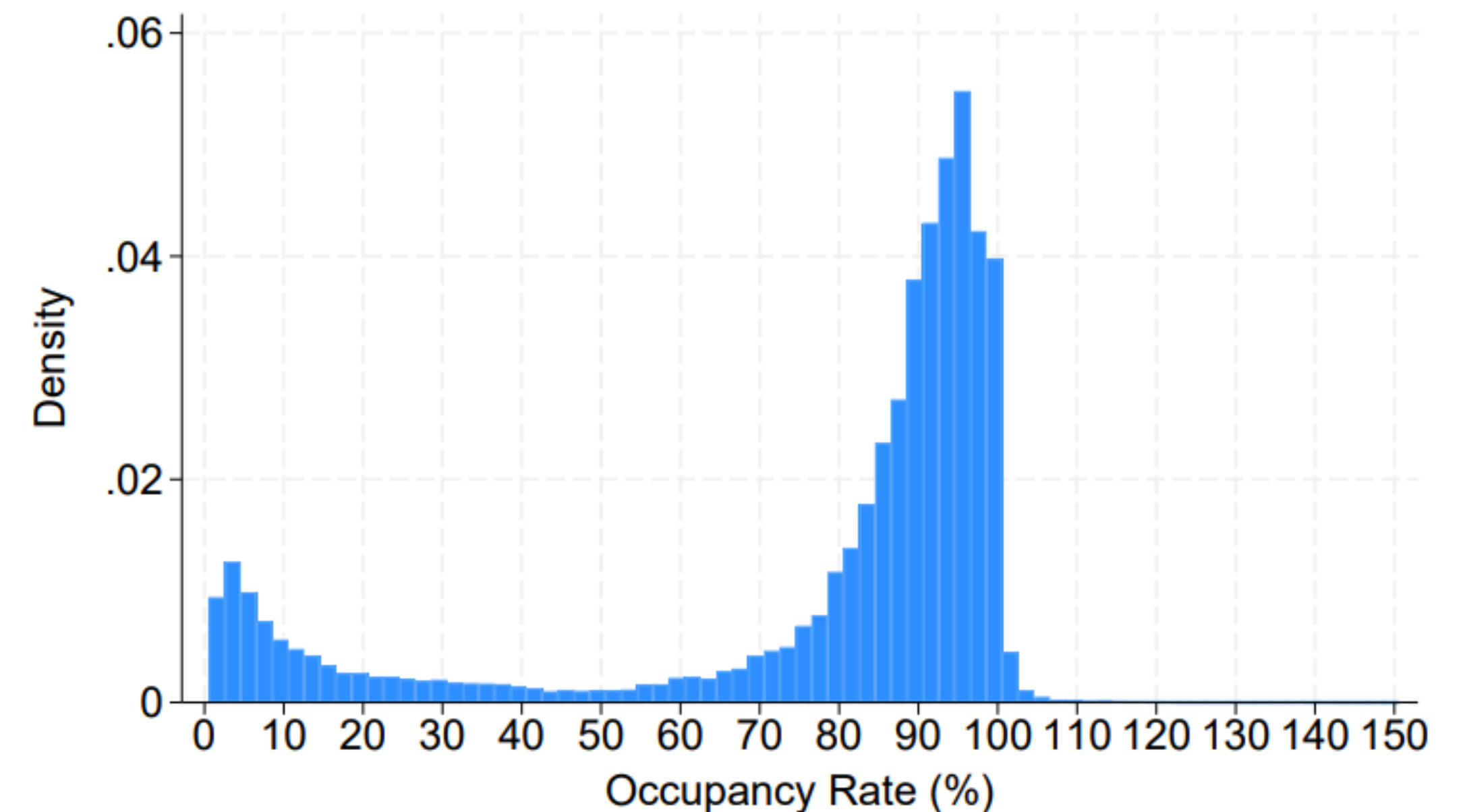
- Data source:
 - Long-term care insurance claims data (similar to medical claims)
 - Survey of Long-Term Care Facilities
- Sample period: April 2011 - March 2018
- Construct **facility-by-date panel data** including
 - Occupancy rates
 - Num. of admissions, discharges, and deaths
 - Patient characteristics

Table 1: Descriptive Statistics

	Mean (1)	SD (2)	p10 (3)	p90 (4)
Facility-date (Obs. = 6,759,473, #Facilities = 3,073)				
Number of beds	83.44	30.45	48	100
Number of physicians	0.72	0.56	0	1
Number of nurses	8.96	3.76	5	13
Number of care staffs	26.02	10.37	13	38
Occupancy rate (pp)	76.73	30.20	12.86	98.15
Care level 1	8.11	6.77	0	17.14
Care level 2	13.74	8.21	1.25	24.00
Care level 3	17.72	9.11	2.50	28.00
Care level 4	20.27	10.25	3.00	32.00
Care level 5	15.70	11.77	2.00	30.00
Number of admissions	0.47	0.85	0	2
Short stay	0.27	0.65	0	1
Long stay	0.20	0.47	0	1
Number of discharges	0.47	0.85	0	2
Short stay	0.27	0.65	0	1
Long stay	0.19	0.49	0	2
Home	0.04	0.21	0	0
Hospital	0.07	0.27	0	0
Death	0.01	0.12	0	0

- Num. of facilities = 3,073
- Mean num. of beds = 83
- Occupancy = 85-100%
- Daily admission & discharge (inpatient)
= 0.2

Figure 1: Distribution of Occupancy



Empirical Strategy

Event Study Design

Dynamic effect of exogenous discharge due to patient deaths on admissions/discharges:

$$Y_{jt} = \sum_{k=-84}^{84} \beta_k Deaths_{jt-k} + \lambda' X_{jt} + \gamma_{jy} + \gamma_t + \varepsilon_{jt} \quad (3)$$

- Y_{jt} : Num. of admissions (live discharges) of facility j on date t
- $Deaths_{jt-k}$: Num. of patients who died at facility j on date $t - k$
(k ranges from -84 to 84)
- X_{jt} : Patient characteristics at facility j on date t
- γ_{jy} : Facility-by-year FEs, γ_t : Date FEs

Y_{jt} and $Deaths_{jt-k}$ are measured as a percentage point (pp) of occupancy

Standard errors are clustered at the municipality level

IV Estimation

Effect of bed occupancy on admissions/discharges:

$$Y_{jw(t)} = \beta OC_{jt} + \lambda' X_{jt} + \gamma_{jy} + \gamma_t + \varepsilon_{jt} \quad (4)$$

- $Y_{jw(t)}$: Num. of admissions (live discharges) at facility j in w weeks after date t
- OC_{jt} : Occupancy rate at facility j on date t

We use patient deaths on the previous day, $Deaths_{jt-1}$, as an IV for OC_{jt}

Response at Different Baseline Occupancy

Estimate IV regression:

1. By occupancy levels (Facility heterogeneity?)

Ex) High-quality facilities have high occupancy and more applicants
(not differential congestion)

2. By facility-specific occupancy percentiles

3. With interaction term:

$$Y_{jw(t)} = \beta_1 OC_{jt} + \beta_2 OC_{jt} \times I \left\{ OC_{jt} \geq L^o \right\} + \lambda_2' X_{jt} + \gamma_{jy} + \gamma_t + \varepsilon_{jt} \quad (5)$$

- $I \left\{ OC_{jt} \geq L^o \right\}$: takes one if occupancy is higher than a certain level L^o
- Use the num. of **two-week patient deaths** as an additional IV

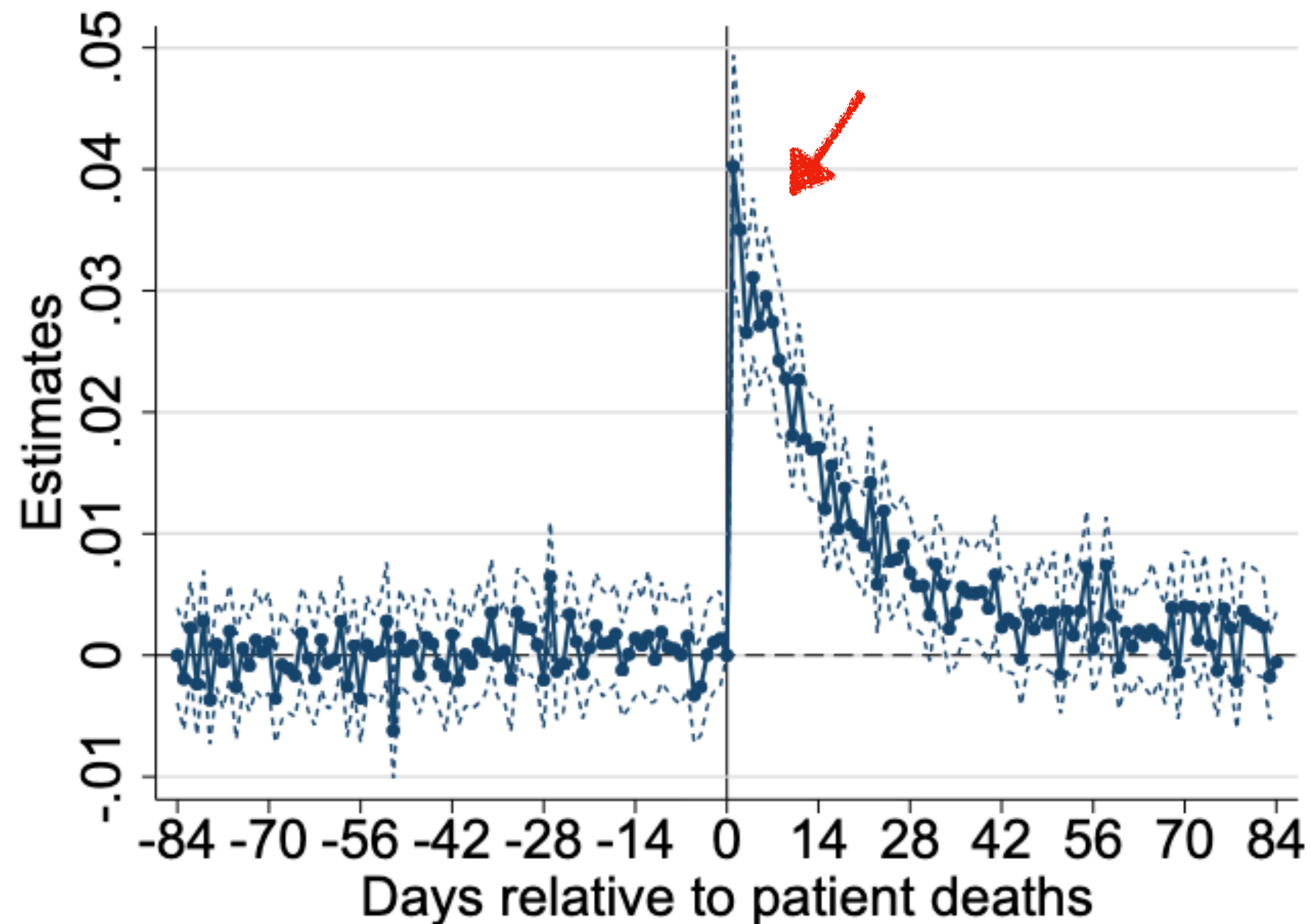
Identification Concerns

- Check pre-trend in event study design
- Patients respond to daily death events? → Unlikely
 - Admission: It is difficult for applicants to gather daily death info
 - Discharge: Patient requires advance planning
- Facility-level health shocks? → Unlikely
 - Trend of hospitalization does not change around the timing of patient deaths

Results

Event Study Estimates (Admission)

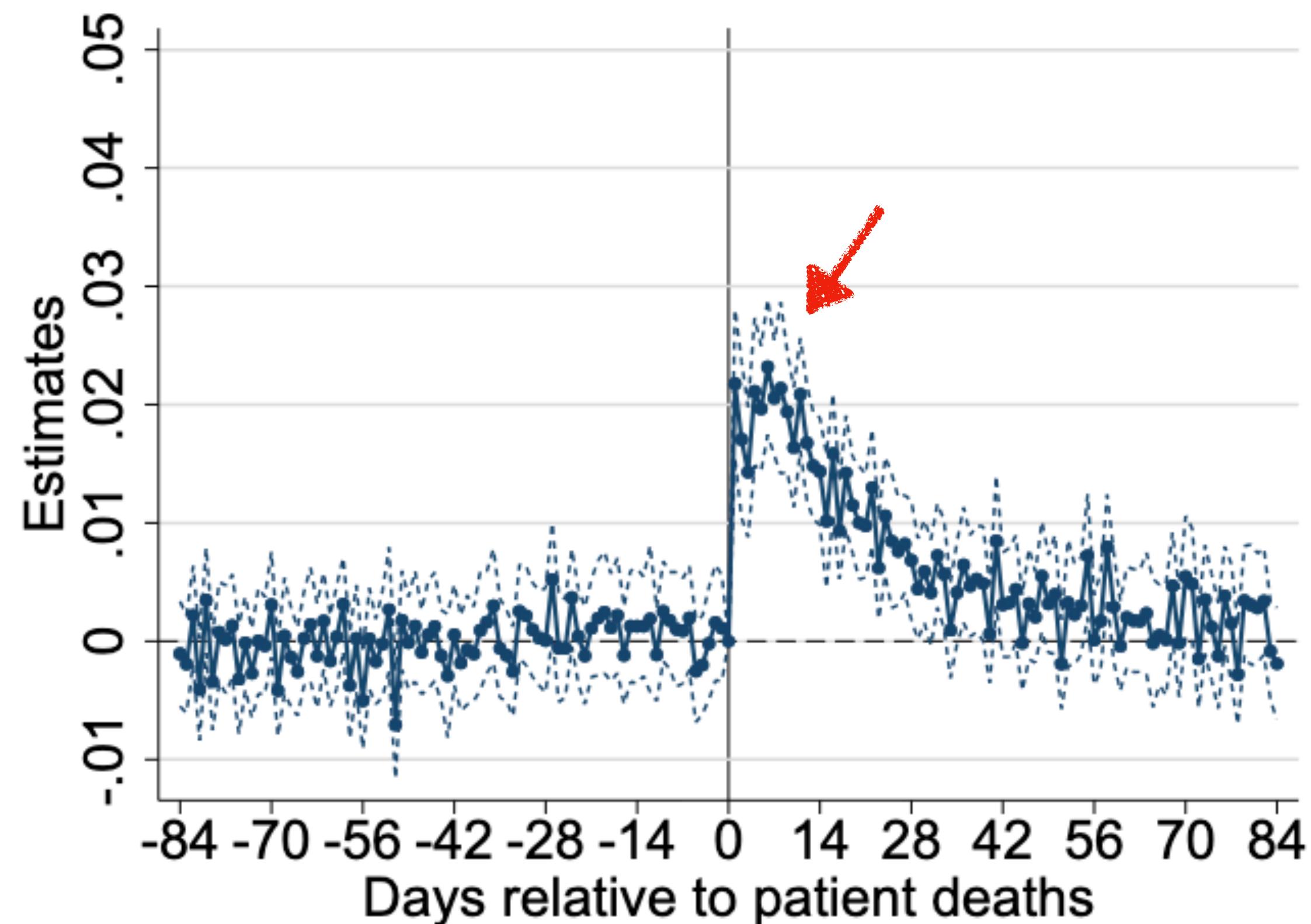
Figure 2: Event Study Estimates
for Admission (Whole sample)



- We set $\beta_0 = 0$ as a reference
- No pre-trend
= The timing of deaths is plausibly exogenous
- Admissions increases right after patient deaths
- Admission response continues about a month
= It takes time to adjust OC

Event Study Estimates (Admission)

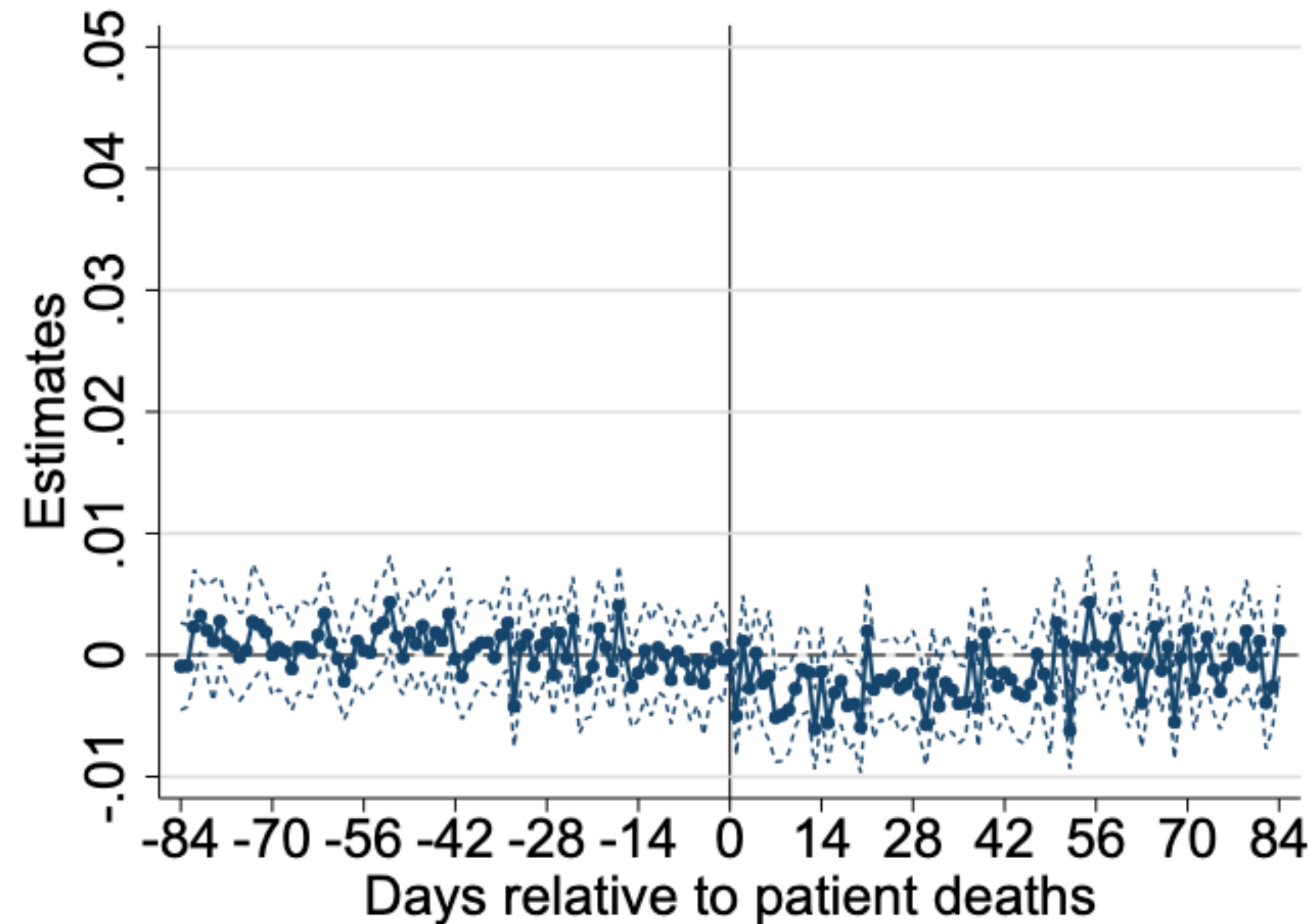
Figure 3: Event Study Estimates
for Admission (Full OC excluded)



- Even if occupancy is not full, admissions respond to patient deaths
- The increase is smaller than whole sample
- Prev. results include mechanical effect of binding capacity constraints

Event Study Estimates (Discharge)

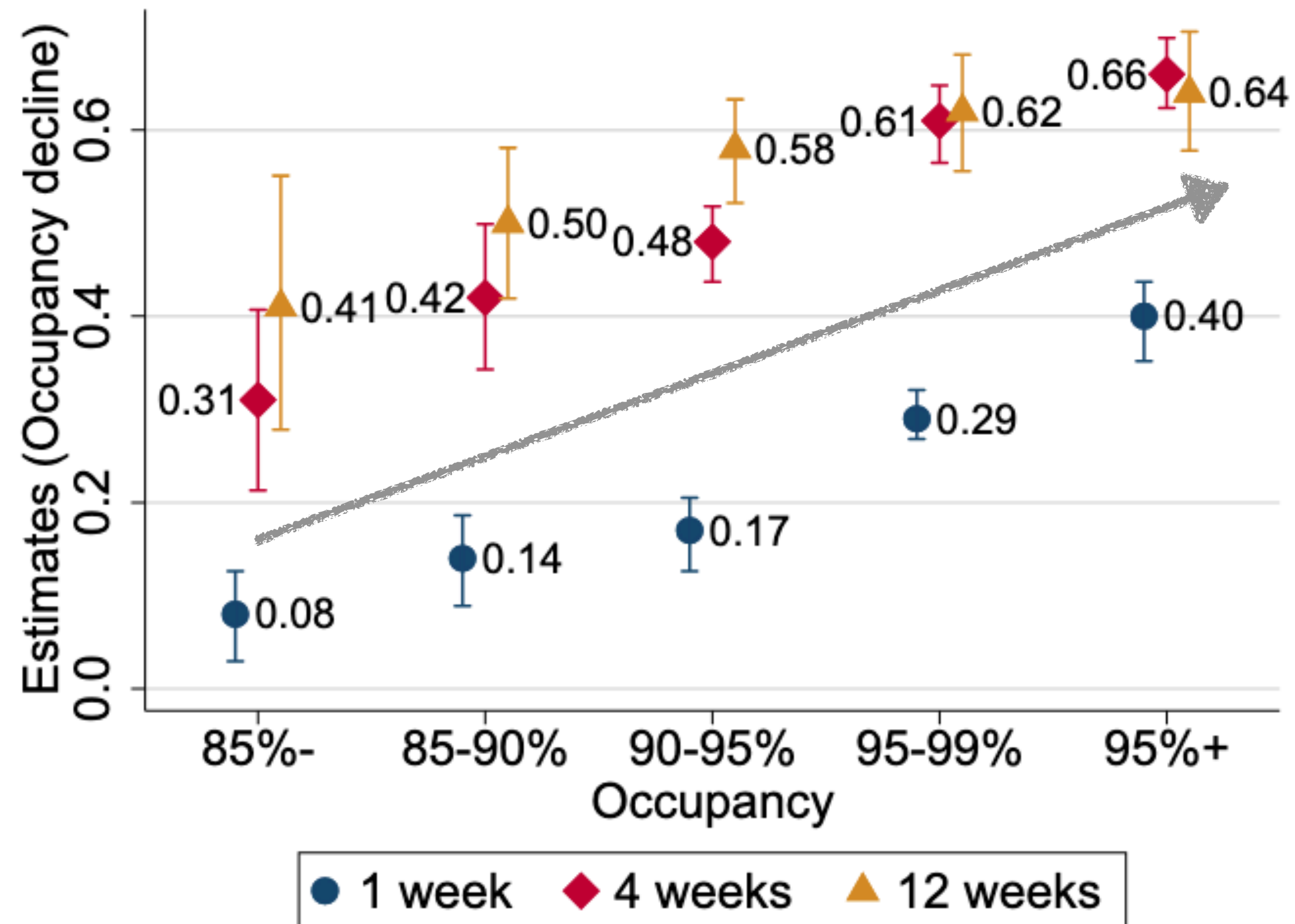
Figure 4: Event Study Estimates
for Discharge (Whole sample)



- No pre-trend
- Discharges slightly decrease after patient deaths, but not significant
- The adjustment cost of discharge might be higher than that of admission (Discharge require advance planning, etc)

IV Estimates by Occupancy Level

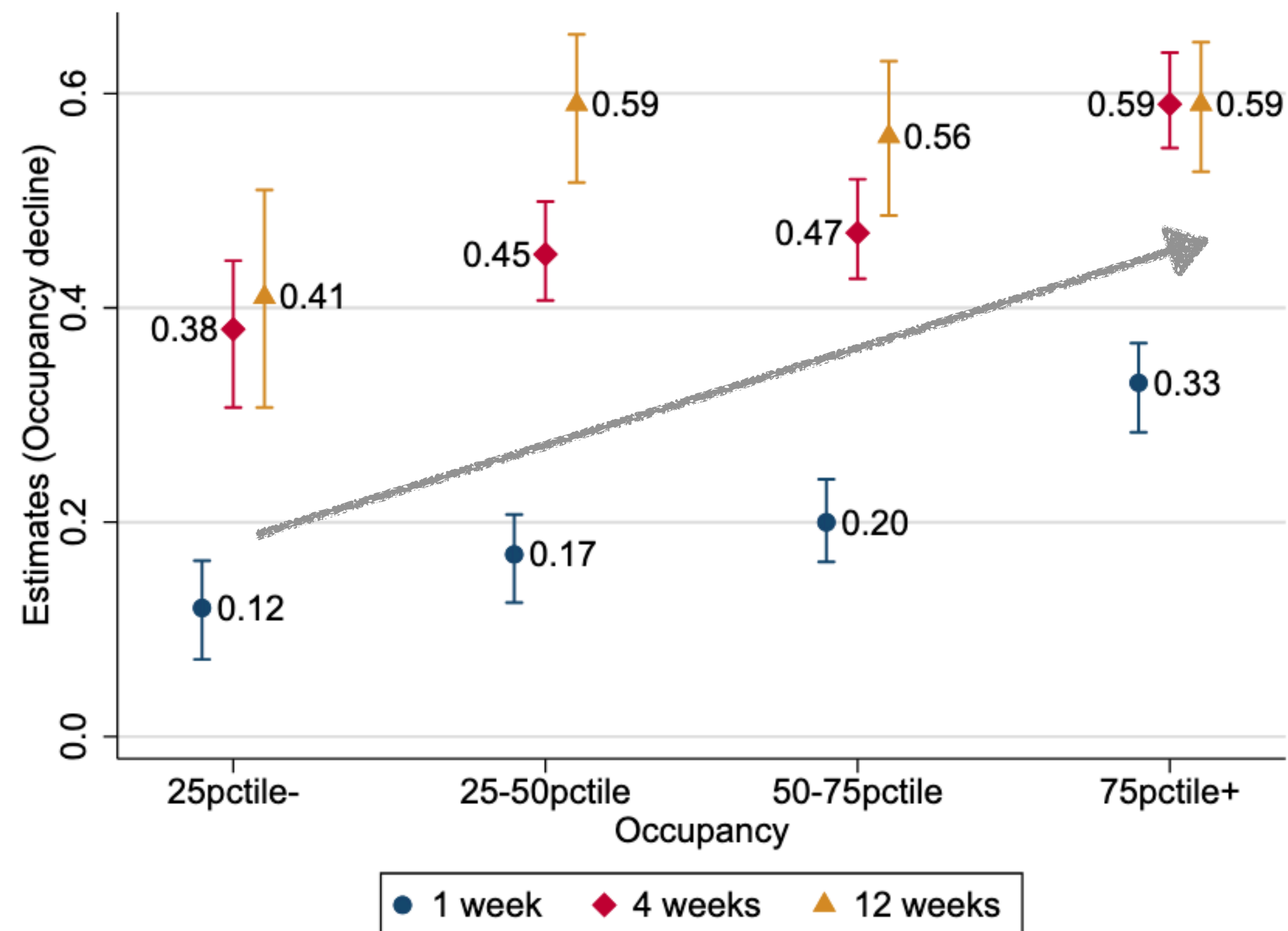
Figure 5: The Effect of Occupancy Decline on Admissions (By OC level)



- Outcomes are admissions in 1, 4, and 12 weeks.
 - At any OC level, the OC decline increases admissions
 - The admission response is large when OC is high
- ⇒ **Cost effect > Income effect**

IV Estimates by Occupancy Percentile

Figure 6: The Effect of Occupancy Decline on Admissions (By OC Percentile)



- Estimate IV regression by facility-specific OC percentile
 - The admission response is large when OC is high
- ⇒ **Cost effect > Income effect**

IV Estimates with Interaction Term

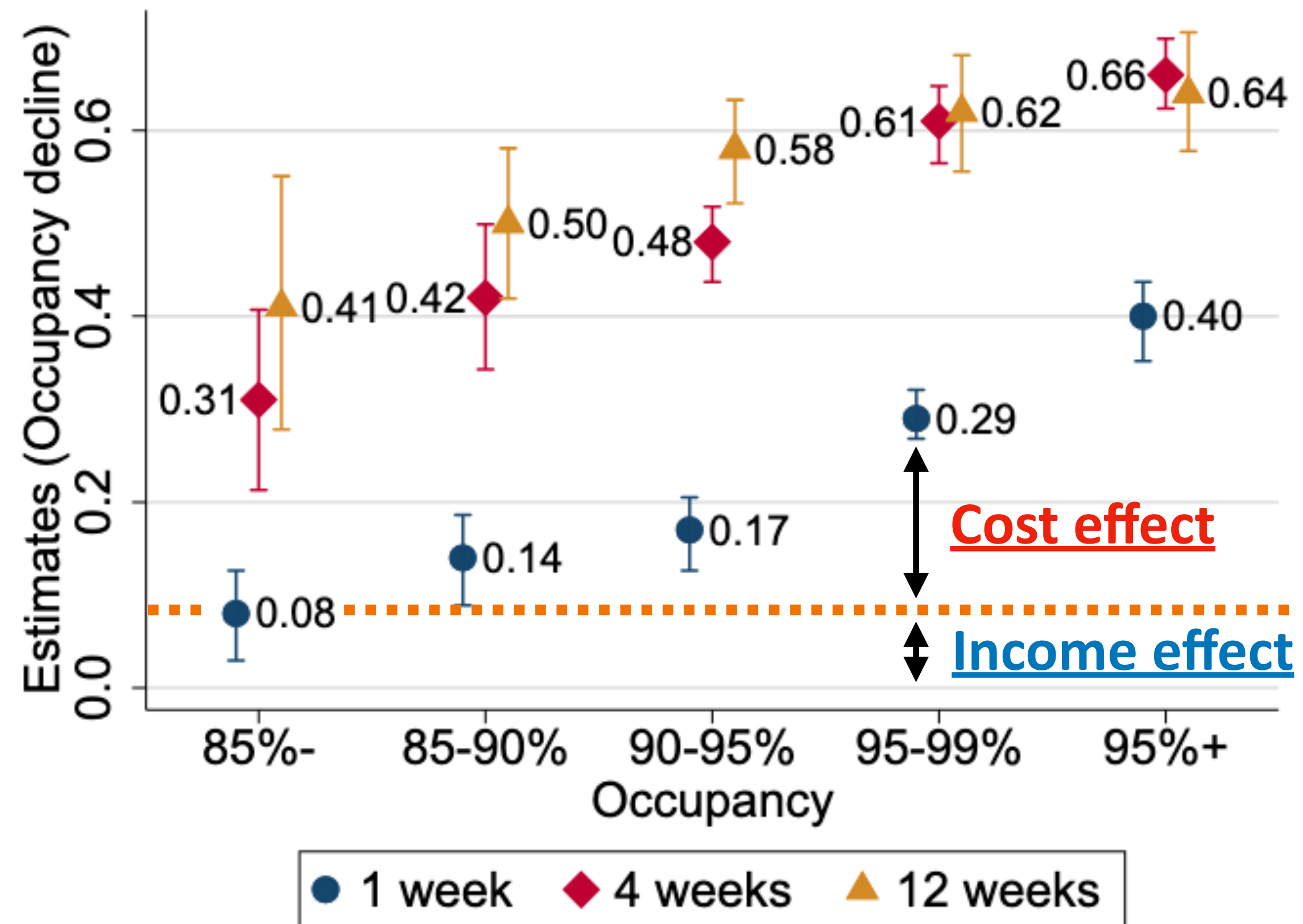
Table 2: IV Estimates (with interaction)

	1-Week Admission			4-Week Admission		
Occupancy (decline)	0.307*** (0.0266)	0.291*** (0.0218)	0.257*** (0.0161)	0.614*** (0.0248)	0.600*** (0.0218)	0.573*** (0.0186)
Occupancy (decline) ×						
I(Occupancy ≥ 95pp)	0.0517*** (0.0129)			0.0424*** (0.0140)		
I(Occupancy ≥ 90pp)		0.0594*** (0.0137)			0.0487*** (0.0159)	
I(Occupancy ≥ 85pp)			0.110*** (0.0270)			0.0904*** (0.0296)
Cragg-Donald F-stats	164.0	141.4	78.05	164.0	141.4	78.05
N	6,673,744	6,673,744	6,673,744	6,673,744	6,673,744	6,673,744

**Cost effect
> Income effect**

How Important Cost Effect?

Figure 5: The Effect of Occupancy Decline on Admissions (By OC level)



- Recall that income effects decrease with occupancy
- If we focus on 1-week admission, income effect is “at most” 0.08
- Ex) OC 95-99%:
Cost effect is “at least” **72.4%** of admission response

Policy Implication

- Which facilities should be allocated additional capacity?
- We showed that cost effect is the main driver of admissions for congested facilities
 - Assume that cost effect is more valuable than income effect in admission
→ Congested facilities should be prioritized for additional capacity
 - Sound obvious? It's not!
- Suppose all admission responses are interpreted as “income effects”
 - Additional capacity is the least valuable for congested facilities
 - They exhibit the largest (potentially wasteful) increase in admission

Conclusion

- In healthcare policy, it is crucial to understand the trade-offs between
 - capacity constraint (congestion)
 - supplier-induced demand
- The relative importance of two mechanisms is testable
- In Japanese nursing facility context, capacity constraint is the main driver of admission response
- Our finding emphasizes the importance of considering capacity constraint as well as supplier-induced demand